# Recitation 6

Correlation, Independence, and Causation
Review and Practice Problems

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*Reading:* *Probability and Statistics for Data Science*, Sections 8.2–8.6 and 8.8.

## 1 Review: covariance and correlation

Assume finite second moments. Write $\mu_{\widetilde{x}} = \mathbb{E}[\widetilde{x}]$ and $\sigma_{\widetilde{x}}^2 = \mathrm{Var}(\widetilde{x})$.

$$\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = \mathbb{E}[(\widetilde{x} - \mu_{\widetilde{x}})(\widetilde{y} - \mu_{\widetilde{y}})] = \mathbb{E}[\widetilde{x}\widetilde{y}] - \mu_{\widetilde{x}}\mu_{\widetilde{y}}.$$

If both variances are positive, the **correlation coefficient** is

$$\boxed{\rho_{\widetilde{x},\widetilde{y}} = \frac{\mathrm{Cov}(\widetilde{x}, \widetilde{y})}{\sigma_{\widetilde{x}}\sigma_{\widetilde{y}}}.}$$

Correlation measures the direction and strength of a *linear* association. It is dimensionless; covariance depends on the units.

**Properties to remember.**

- **Symmetry and bounds:** $\rho_{\widetilde{x},\widetilde{y}} = \rho_{\widetilde{y},\widetilde{x}}$ and $-1 \le \rho_{\widetilde{x},\widetilde{y}} \le 1$.
- **Shifts and rescaling:** for nonzero constants $b, d$,

  $$\mathrm{Cov}(b\widetilde{x} + a, d\widetilde{y} + c) = bd\ \mathrm{Cov}(\widetilde{x}, \widetilde{y}), \qquad \rho_{b\widetilde{x}+a,d\widetilde{y}+c} = \mathrm{sign}(bd)\rho_{\widetilde{x},\widetilde{y}}.$$

  Positive changes of scale leave correlation unchanged; reversing one variable's sign reverses the correlation.
- **Perfect correlation:** $|\rho| = 1$ exactly when $\widetilde{y} = \alpha + \beta\widetilde{x}$ with probability one, for some $\beta \neq 0$. The sign of $\rho$ is the sign of $\beta$.
- **Variance of a sum:** $\mathrm{Var}(\widetilde{x} + \widetilde{y}) = \mathrm{Var}(\widetilde{x}) + \mathrm{Var}(\widetilde{y}) + 2\,\mathrm{Cov}(\widetilde{x}, \widetilde{y})$.

**Connection to linear prediction.** The best *affine* predictor of $\widetilde{y}$ from $\widetilde{x} = x$ is

$$\ell_{\mathrm{MMSE}}(x) = \mu_{\widetilde{y}} + \frac{\mathrm{Cov}(\widetilde{x}, \widetilde{y})}{\mathrm{Var}(\widetilde{x})}(x - \mu_{\widetilde{x}}) = \mu_{\widetilde{y}} + \rho_{\widetilde{x},\widetilde{y}}\frac{\sigma_{\widetilde{y}}}{\sigma_{\widetilde{x}}}(x - \mu_{\widetilde{x}}).$$

Its MSE is $(1 - \rho_{\widetilde{x},\widetilde{y}}^2)\sigma_{\widetilde{y}}^2$. Thus $\rho^2$ is the fraction of variance explained by the best affine predictor. If $\rho = 0$, that predictor is just the mean; nonlinear prediction may still help. A constant variable has zero covariance with every variable, but its correlation coefficient is *undefined*.

## 2 Mini-lecture: uncorrelated versus independent

**Uncorrelated** means $\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0$. **Independent** means that, for every pair of events defined by the two variables,

$$\mathbb{P}(\widetilde{x} \in A, \widetilde{y} \in B) = \mathbb{P}(\widetilde{x} \in A)\mathbb{P}(\widetilde{y} \in B).$$

For discrete variables this is equivalent to $p_{\widetilde{x},\widetilde{y}}(x, y) = p_{\widetilde{x}}(x)p_{\widetilde{y}}(y)$ for every $x, y$.

**Independence implies uncorrelation.** With finite second moments, independence gives

$$\mathbb{E}[\widetilde{x}\widetilde{y}] = \mathbb{E}[\widetilde{x}]\mathbb{E}[\widetilde{y}] \quad \Longrightarrow \quad \mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0.$$

The reverse implication is false: covariance checks only one mixed moment, whereas independence constrains the entire joint distribution.

**Worked example: a nonlinear relationship with zero correlation.** Let $\widetilde{x}$ take $-1, 0, 1$ with equal probability, and set $\widetilde{y} = \widetilde{x}^2$. The possible pairs are

| $(x, y)$ | $(-1, 1)$ | $(0, 0)$ | $(1, 1)$ |
|---|---|---|---|
| $\mathbb{P}(\widetilde{x} = x, \widetilde{y} = y)$ | $1/3$ | $1/3$ | $1/3$ |

By symmetry,

$$\mathbb{E}[\widetilde{x}] = 0, \qquad \mathbb{E}[\widetilde{y}] = \frac{2}{3}, \qquad \mathbb{E}[\widetilde{x}\widetilde{y}] = \frac{-1 + 0 + 1}{3} = 0.$$

Hence $\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0$, and $\rho_{\widetilde{x},\widetilde{y}} = 0$ because both variables have positive variance. However,

$$\mathbb{P}(\widetilde{x} = 0, \widetilde{y} = 0) = \frac{1}{3} \neq \frac{1}{9} = \mathbb{P}(\widetilde{x} = 0)\mathbb{P}(\widetilde{y} = 0).$$

They are dependent: observing $\widetilde{x}$ determines $\widetilde{y}$ exactly. The positive and negative contributions to covariance cancel.

**The Gaussian exception.** If $(\widetilde{x}, \widetilde{y})$ is *jointly Gaussian*, then uncorrelated $\Longleftrightarrow$ independent. It is not enough that each variable separately has a Gaussian distribution.

## 3 Review: correlation versus causation

Association describes how variables vary together. A causal question asks what would happen if we *intervened* to change one variable. For example, rain can cause both umbrella use and wet sidewalks:

$$\text{Umbrella use} \longleftarrow \text{Rain} \longrightarrow \text{Wet sidewalks}.$$

Their association does not establish that using an umbrella makes a sidewalk wet. A common cause is a **confounder**; reverse causation can also produce association. Randomized assignment can support a causal comparison. Observational adjustment requires additional causal assumptions, as in Recitation 5. Neither a large $|\rho|$ nor a small $|\rho|$ alone determines a causal effect; "variance explained" is a predictive statement.

## Practice problems

Three core problems; Problem 4 is a backup for a fast class. The indicated times are for independent work before discussion.

### Problem 1: Covariance properties (Exercise 8.2)

*Independent work: 10 minutes.* Let $\widetilde{a}$ and $\widetilde{b}$ have finite variances. Prove the following:

(a) For any constants $\alpha, \beta \in \mathbb{R}$,

$$\operatorname{Cov}(\beta\widetilde{a} + \alpha, \widetilde{b}) = \beta \operatorname{Cov}(\widetilde{a}, \widetilde{b}).$$

Conclude that if $\widetilde{a}$ and $\widetilde{b}$ are uncorrelated, so are $\beta\widetilde{a} + \alpha$ and $\widetilde{b}$.

(b) The covariance satisfies

$$-\sqrt{\operatorname{Var}(\widetilde{a})\operatorname{Var}(\widetilde{b})} \le \operatorname{Cov}(\widetilde{a}, \widetilde{b}) \le \sqrt{\operatorname{Var}(\widetilde{a})\operatorname{Var}(\widetilde{b})}.$$

Include the case where one variance is zero.

(c) The sum and difference satisfy

$$\operatorname{Cov}(\widetilde{a} + \widetilde{b}, \widetilde{a} - \widetilde{b}) = \operatorname{Var}(\widetilde{a}) - \operatorname{Var}(\widetilde{b}).$$

### Problem 2: An affine function (Exercise 8.11)

*Independent work: 7 minutes.* The random variables $\widetilde{a}$ and $\widetilde{b}$ satisfy

$$\mu_{\widetilde{a}} = 2, \qquad \mu_{\widetilde{b}} = -2, \qquad \sigma^2_{\widetilde{a}} = 4, \qquad \sigma^2_{\widetilde{b}} = 9, \qquad \rho_{\widetilde{a},\widetilde{b}} = 0.25.$$

Define $\widetilde{c} = 2\widetilde{b} + 3$. Find the linear minimum mean-squared-error predictor of $\widetilde{c}$ given $\widetilde{a} = a$.

*Suggested steps:* Convert correlation to covariance; find $\mathbb{E}[\widetilde{c}]$ and $\mathrm{Cov}(\widetilde{a}, \widetilde{c})$; then identify the slope and intercept of the predictor.

### Problem 3: Ice cream sales (Exercise 8.15)

*Independent work: 8 minutes.* Alma is analyzing data for an ice cream company. The marketing team argues that a strong correlation between advertising expenses and sales shows that its advertising is effective. Alma obtains the following *sample covariance matrix*:

| | Advertising | Sales | Temperature |
|---|---|---|---|
| Advertising | 900 | 960 | 540 |
| Sales | 960 | 1600 | 720 |
| Temperature | 540 | 720 | 400 |

(a) Compute the sample correlation between advertising expenses and sales.

(b) Should Alma accept the team's conclusion? Use the temperature entries to inform your explanation, and distinguish association from a causal effect.

*Reminder:* Divide a sample covariance by the product of the sample standard deviations. The diagonal entries above are variances.

### Problem 4 (backup): A point in a disk

*Independent work: 5 minutes; use only if the class is ahead.* Choose a point uniformly from the unit disk. Its coordinates have joint pdf

$$f_{\widetilde{x},\widetilde{y}}(x,y) = \begin{cases} 1/\pi, & x^2 + y^2 \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Without evaluating integrals, use symmetry to find $\mathbb{E}[\widetilde{x}]$, $\mathbb{E}[\widetilde{y}]$, and $\mathbb{E}[\widetilde{x}\widetilde{y}]$. Are the coordinates uncorrelated?

(b) Are the coordinates independent? Justify your answer using the disk's support or a pair of events.

*Source:* adapted from the 2025 archive, `Recitations/Recitation 1 Problems.pdf`, Problem 2 [3.31]. The covariance question has been added.